

20 DOLLARS OF ICE JUGS KEEPS A FREEZER COLD: EVERY NUMBER

Jugs of ice fill the empty space, so a half empty freezer starts acting like a full one. It beats a 1,200 dollar generator that still needs fuel. Two days is a planning number, not a promise.

WHAT IT IS

After the power dies, a half empty freezer holds for about a day. Twenty dollars of frozen water jugs should stretch that toward two. The jugs fill the air space, so the freezer behaves like a packed one.

about 48 hr	The Department of Agriculture says a full freezer holds its temperature about 48 hours with the door shut.
about 24 hr	A half full one holds about 24 hours. Jugs of ice close that gap.
about 20 dollars	Fourteen one gallon jugs of water. Tap water in jugs you already own costs nothing.

No test of this exact setup was found, so the two day figure is an estimate. It rests on the physics and the federal numbers, and federal guidance already says to freeze containers of water.

WHAT THE GENERATOR REALLY COSTS

The rival is one well known model, Honda's 2,200 watt inverter.

about 1,200 dollars	Price of the Honda, which also needs fuel the whole time. Cheaper generators and battery stations exist, but they still need fuel or a recharge when the grid is down.
up to 8.1 hr	Honda's rated run time per tank at a quarter load. The tank holds just under a gallon.
about 25 W	A freezer averages far less than a quarter load, about 25 watts across a year for an efficient chest model. The compressor draws more while running and needs a surge to start.
about 1 in 60	Twenty dollars of jugs is about one sixtieth of the generator's price. Honda publishes no freezer run time; a tank would probably last longer than 8 hours, but two days still means refilling it.

Finding gas can be the harder part. After Hurricane Ida in 2021, GasBuddy counted more than half the gas stations in New Orleans and Baton Rouge with no fuel. Nine days after the storm, that was still true.

IF YOU DO OWN A GENERATOR

- ☐ Run it outdoors only, at least 20 feet from the house. Point the exhaust away from windows, doors, and vents. A garage with the door open still counts as partly enclosed.
- ☐ Put carbon monoxide alarms with batteries on every level of the house and outside each bedroom. The Consumer Product Safety Commission counts roughly 85 to 100 carbon monoxide deaths a year from portable generators.
- ☐ Shut the generator off and let it cool before you refuel it.

WHAT IS AT STAKE IN THE FREEZER

about 35 lb	Makers use a rule of thumb of about 35 pounds of food per cubic foot.
about 500 lb	So a packed 15 cubic foot freezer holds roughly 500 pounds.
6 dollars 92	Federal average price of a pound of ground beef. Say 100 pounds of the 500 is beef; that is about 690 dollars of beef, and the rest is on top.

THE SPOILAGE CLOCK

40 F, 2 hr	Perishable food above 40 degrees Fahrenheit for two hours or more goes in the trash. It is one hour if the room is above 90.
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Your nose cannot sort out which food is safe. The University of Missouri Extension warns that thawed meat and vegetables can spoil without any distinct odor.

WHY EMPTY SPACE IS THE PROBLEM

One woodworker logged the temperature inside a 14 cubic foot upright freezer in a basement at about 66 degrees.

3.5 hr	Empty, it warmed from 0 to 32 degrees in just 3 and a half hours.
about 26 hr	His projection for a freezer less than half full.
about 36 hr	His projection for a packed one. These are projections from one freezer, not outage tests. A packed one at 36 is also shorter than the federal 48, so freezers vary.
5 s = 7 min	In that upright, opening the door for 5 seconds let in as much heat as about 7 minutes of normal warming. Cold air is heavier, so it pours out of an upright's front. A chest freezer spills less. Every look still costs you.
under 3 Wh	All the air in a 15 cubic foot freezer weighs about a pound; warming it from 0 to 32 takes under 3 watt hours. An empty freezer has very little stored cold to spend.

The room matters as much as the freezer. Heat leaks in faster from a hot garage than a cool basement, so two days is a planning number, not a promise. In a garage, the jugs matter more in summer.

WHY A JUG OF ICE HOLDS SO MUCH

334 J	It takes 334 joules to melt a single gram of ice. The whole time it melts, the ice stays at 32 degrees, because the energy breaks the crystals apart instead of raising the temperature.
7.5 lb	A gallon jug filled to about 90 percent holds 7 and a half pounds of ice.
about 350 Wh	Melting that soaks up about 315 watt hours; warming the ice from 0 to 32 first adds about 35, so call it 350 watt hours of cold per jug.

Food freezes a few degrees below 32, thanks to dissolved salt and sugar, so near 32 some food starts to soften. That is still under 40, but ice in a jug does not prove every package is cold enough. Check the food itself.

HOW MANY JUGS

3 to 4	Jugs per empty cubic foot roughly matches the food they replace, since a jug stores more cold than the same weight of food.
about 25	A half empty 15 cubic foot freezer has about 7 and a half cubic feet of space, around 25 jugs.
14	Twenty dollars buys 14 jugs, about half. Fill the rest with tap water in clean bottles you already have. Each jug past the first 14 adds another 350 watt hours, and most cost nothing.

Milk jugs and juice bottles both work. Check each one for cracks and leaks first, because old thin plastic can split when the water freezes.

BUILDING IT BEFORE THE STORM

- 1 A day before a forecast storm, turn the freezer to its coldest setting. Turn it back afterward.
- 2 Use sturdy plastic containers made for freezing. Skip glass jars, which can crack at the neck as water expands.
- 3 Water grows about 9 percent when it freezes, so fill each container about 90 percent. For a store bought gallon, pour out a cup and a half, then cap it.
- 4 Add the jugs a few at a time, upright. One maker limits new food to about 3 pounds per cubic foot per day, so for a 15 cubic foot freezer that is about 5 or 6 jugs a day.
- 5 A gallon can take a day or two to freeze all the way through. Slush in the middle means less cold stored. Tape the date you started freezing on each jug.
- 6 Group the food together in the middle, and fill the gaps around it and on top with jugs. Use small bottles for small gaps. Do not overload the shelves, and make sure the door or lid still seals.
- 7 In a chest freezer, stand jugs along the walls and lay a layer across the top. In an upright, give every shelf a row and keep jugs off the door seal.

THERMOMETER AND THE TWO CHECKS

- Keep the freezer at 0 degrees Fahrenheit or below. Pick a thermometer with a wire probe and an outside display so you can read it without opening the lid. Budget a few dollars on top of the 20.
- Freeze a cup of water solid and lay a coin on top. If you come home and the coin has sunk to the bottom, the freezer thawed and refroze while you were gone.
- Frost free upright: keep jugs and food clear of the inside air vents. Whirlpool and Maytag both say blocked vents cause temperature problems.

YOUR FIRST TWO DAYS

- 1** When the power dies, write down the time. Then leave the door or lid alone.
- 2** Read the outside display a few times a day, and jot each reading and time on paper by the freezer. A slow creep up is normal; a climb toward 32 is your signal to line up more cold.
- 3** Around the 24 hour mark, decide your next move if power is still out: a friend with freezer space, or price block ice or dry ice now while stores have it.
- 4** Use the most perishable food first. If you have a cooler, pack it with block ice and move the food you would hate to lose.

The blanket is optional. The University of Missouri Extension says to cover the freezer with blankets to hold in the cold, pinned away from the vents. No insulation rating was found; a Harbor Freight moving blanket lists for about 10 dollars. Use it only while you are home, keep it off every vent, and take it off before the power comes back.

Melted jug water that stayed in a clean, capped container is drinking water. That is worth having when the taps are off too.

WHEN THE POWER COMES BACK

- ☐ Remove the blanket first, then check the food.
- ☐ Food that still has ice crystals, or reads 40 degrees Fahrenheit or below, can be refrozen or cooked. Throw out thawed ice cream.
- ☐ Perishable food above 40 for more than two hours goes in the trash. Do not wait for it to reach room temperature.
- ☐ Never taste food to decide, because the bacteria that make people sick do not change the smell or the look. Throw out anything you are unsure about.

IF THE OUTAGE RUNS PAST A DAY OR TWO

Do not wait for a fixed deadline. Watch the thermometer and arrange more cold while your food is still safe. Once the last ice melts, the real cold is gone.

50 lb, 2 days

Fifty pounds of dry ice should keep an 18 cubic foot, fully stocked freezer cold for 2 days, per the Food and Drug Administration.

3 to 4 days

The University of Missouri Extension gives 3 to 4 days for a full 20 cubic foot freezer. These cover different freezers, and neither is a guarantee.

DRY ICE SAFETY

- ☐ **Wear insulated gloves made for extreme cold, and eye protection.** Dry ice burns skin.
- ☐ **Use it only in a well ventilated room,** since it turns to carbon dioxide gas as it warms. Never seal it in an airtight container.
- ☐ Set it on a board or heavy cardboard above the food, not directly on it.
- ☐ Cooking cannot rescue food that has gone bad. Some bacteria leave toxins that cooking does not destroy.

THE FRIDGE RUNS A SHORTER CLOCK

The Department of Agriculture says a refrigerator holds a safe temperature for about 4 hours with the door closed. Jugs help there too, and federal guidance says to freeze containers of water for the fridge as well.

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Sketchy Survival 101. Physics first survival. Every number on this sheet comes straight from the video, with its hedges kept: projections from one freezer, single maker claims, dated storm figures, and estimates that are not guarantees. Two days is a planning number, not a promise. Watch your own thermometer before you act.